LESSON 6.2
USING MODELS TO REPRESENT RATIOS

LESSON INTRODUCTION

MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or a:b where $b \neq 0$.

LEARNING TARGETS

- I can write and interpret ratios.
- I can represent ratios using various models.

BENCHMARK COHERENCE

BUILDING ON

MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How do you write ratios?
- How do you interpret ratios?
- How do you represent ratios using models?

LESSON NARRATIVE

In this lesson, students begin by describing and comparing ratios of cups of milk to scoops of chocolate powder. They explore representing a ratio with a double number line and use it to interpret the ratio. Students make the connection between representing ratios using double number lines and tape diagrams, interpreting ratios through these visual representations.

COMPONENT SUMMARY

6.2.1 WARM-UP *(~5 MIN.)*

Describe and compare ratios of cups of milk to scoops of chocolate powder

6.2.2 EXPLORATION *(15 MIN.)*

Explore representing a ratio with a double number line and use it to interpret the ratio

6.2.3 TRY IT *(5-10 MIN.)*

Practice using double number lines to represent and interpret ratios

6.2.4 GUIDED INSTRUCTION *(10 MIN.)*

Students use tape diagrams to represent ratios

6.2.5 YOUR TURN! *(10 MIN.)*

Practice using double number lines and tape diagrams to represent and interpret ratios

6.2.6 WRAP-UP *(~5 MIN.)*

Students use a tape diagram to create a double number line and interpret a ratio

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Ratio

MATERIALS

- Clothesline Number Lines

ADDITIONAL PREPARATION

- Warm-Up 6.2.1 contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may forget that ratios can be expressed three ways: a to b, a:b, or $\frac{a}{b}$.
- Students may forget to simplify ratios, especially when the ratio is not expressed using a fraction bar.
- Students may be confused when they see the same number in different positions in a double number line.

THINGS TO CONSIDER

- Create an anchor chart and allow students to create reference cards with instructions and examples to aid them in writing ratios, interpreting ratios, and representing ratios using various models.
- Utilize the Warm-Up and Exploration as one cohesive, student-centered discovery of double number lines in real-world context.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

6.2.1 Warm-Up and the 6.2.2 Exploration can be structured with the Notice and Wonder routine. Ask students what they notice and wonder about the difference between the ratio of chocolate powder to milk in the recipe and in Dan's mixture. Ask students to share and justify the reasoning behind their responses.

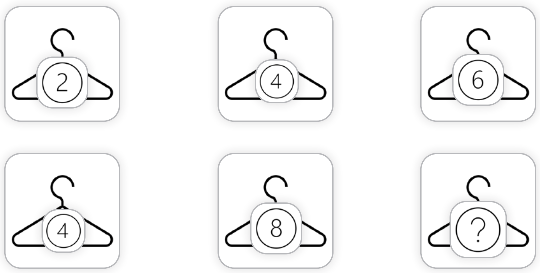
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Throughout the lesson, students see ratios represented in multiple ways. The ratios are expressed three ways: a to b, a:b, or $\frac{a}{b}$. The ratios are represented using tables, double number lines, and tape diagrams. Students draw connections among these representations throughout the lesson, deepening conceptual understanding.

DIFFERENTIATE INSTRUCTION

Clothesline Number Lines offer manipulatives for students to interact with double number lines for hands-on learning. Consider using a kinesthetic representation of the double number lines on clotheslines that cross the room. Students can place and adjust numbers on folded paper/cardstock on the clothesline, either folding pieces of paper or using clothespins to hold each number in place.



6.2.1 WARM-UP: NANA’S CHOCOLATE MILK

Component Overview: *Play the video for the class, then students work independently to describe and compare ratios of cups of milk to scoops of chocolate powder. Students can move on to 6.2.2 Exploration without getting a right answer to question 1b, as they will continue to work on this question in the next component.*

ENRICHMENT

Students may initially respond with “Adding more milk” or “Adding more milk and more chocolate” – prompt them to use a ratio to find out how much more milk and/or chocolate.

Notice 6.2.2 Exploration continues this concept with a double number line.

6.2.1 Warm-Up: Nana’s Chocolate Milk

1. **Dan’s Nana prefers her chocolate milk made with just the right “chocolateyness,” but he made a mistake when mixing it for her. The image below shows what Dan should have mixed and what he actually mixed.**

   a. **Describe what Dan did wrong.**

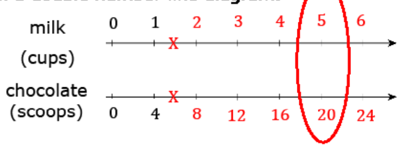
   Dan added one extra scoop of chocolate.

   b. **How could Dan fix it so that the glass contains Nana’s preferred balance of milk and chocolate?**

   Answers will vary. Add ¼ cup of milk.

6.2.2 Exploration: Using Double Line Diagrams to Represent Ratios

Nana's chocolate milk recipe calls for 4 scoops of chocolate for every 1 cup of milk. One way to represent the recipe is with a double number line diagram.



1. **Label each tick mark on the double number line diagram to represent the relationship between cups of milk and chocolate scoops.**
   Shown on diagram.

2. **How does the double number line show the ratio of Nana's preferred balance of milk and chocolate?**
   The double line diagram shows that for every 1 cup of milk there are 4 scoops of chocolate.

3. **How many scoops of chocolate need to be mixed with 5 cups of milk to maintain the same ratio of milk to chocolate from Nana's recipe?**
   For 1 batch of Nana's chocolate mix, Dan needs 1 cup of milk and 4 scoops of chocolate. For 5 times that number of cups, Dan would need 5 times the number of scoops. Therefore, 4 scoops times 5 equals 20 scoops of chocolate.

4. **Circle where the answer to the previous question is represented on the double number line diagram.**
   Shown on diagram.

6.2.2 EXPLORATION: USING DOUBLE LINE DIAGRAMS TO REPRESENT RATIOS

Component Overview: *Students explore representing a ratio with a double number line and use it to interpret the ratio. Consider pairing students to encourage mathematical discourse throughout this component.*

ENRICHMENT

Students should use the scaffolded prompts to explore the connection between the double number line and the ratio that represents the amount of milk needed. Students could also determine the fraction by estimating.

- What is halfway between 4 and 8?
- What is halfway between that? (5 scoops)
- What is halfway between 1 and 2?
- What about halfway between 1 and 1.5?

6.2.2 EXPLORATION: USING DOUBLE LINE DIAGRAMS TO REPRESENT RATIOS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider having students add to their reference card from Lesson 1 with an explanation on how to write ratios, interpret ratios, and represent ratios using various models. Encourage students to include at least one example of each process.Allow students to use their reference cards during the lesson.

TEACHER MOVES

Notice and Wonder

Mathematical discourse during this exploration should sound informal and explorative. Students have not yet connected ratios to double number lines, so as they explore this connection, encourage them to verbalize what they notice about the double number line and how it connects to ratios. Encourage students to work with their partner to verbalize their responses before putting them in writing.

QUESTIONING

Consider asking students who finish early to represent the ratio on the double number line using the three ways they have already learned in the previous lesson. Challenge students to find other equivalent ratios for Nana's chocolate milk recipe. What if you wanted to make chocolate milk for the whole class?

Each pair of numbers on the double number line diagram represents an equivalent ratio. For example, the ratio $6 : 24$ is equivalent to the ratio $2 : 8$.

5. **Complete the statement to interpret these two equivalent ratios in the context of Nana's recipe.**

   If 6 cups of milk are mixed with 24 scoops of chocolate, it will taste the same as mixing 2 cups of milk with 8 scoops of chocolate.

6. **Recall Dan's error when mixing Nana's chocolate milk.**

Nana's Recipe	What Dan Accidentally Mixed
1 cup milk 4 scoops chocolate	1 cup milk 5 scoops chocolate

   a. **Sabrina said Dan's mistake can be fixed by adding 0.5 cup of milk and 1 scoop of chocolate to the glass. Discuss with your partner whether or not you think Sabrina is correct.**

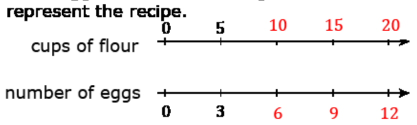
   b. **Complete the statement.**
   If Dan follows Sabrina's recommendation, the glass will contain 1.5 cups of milk and 6 scoops of chocolate.

7. **Use the double number line diagram to determine whether Sabrina's recommendation will result in chocolate milk with Nana's preferred taste.**

   If Sabrina's recommendation will work, mark the locations of this mixture on the double number line diagram with two X's, one on each number line.
   Shown on diagram.

6.2.3 Try It: Using Double Line Diagrams to Represent and Interpret Ratios

1. A recipe for 1 batch of cookies calls for 5 cups of flour and 3 eggs. A double line diagram is shown to represent the recipe.



   a. What is the ratio of flour to eggs in 1 batch of cookies?
   5 *to* 3, 5:3 *or* $\frac{5}{3}$

   b. Complete the diagram to show the amount of flour and eggs needed for 2, 3, and 4 batches of cookies.
   Shown on diagram.

   c. How many eggs and cups of flour are used in 4 batches of cookies?
   12 *eggs and* 20 *cups of flour*

   d. Complete the ratio comparing cups of flour to 6 eggs.
   $\boxed{10} : 6$

   e. Complete the ratio of eggs to 15 cups of flour.
   $\frac{\boxed{9}}{15}$

6.2.3 TRY IT: USING DOUBLE LINE DIAGRAMS TO REPRESENT AND INTERPRET RATIOS

Component Overview: *Students practice using double number lines to represent and interpret ratios. The ratios are expressed three ways: a to b, a:b, or* $\frac{a}{b}$. *The ratios are represented using tables, double number lines, and tape diagrams.*

DIFFERENTIATE INSTRUCTION

MA.K12.MTR.2.1: Demonstrate understanding by representing problems in multiple ways.

Students see ratios represented multiple ways. The ratios are expressed three ways: a to b, a:b, or $\frac{a}{b}$. The ratios are represented using tables, double number lines, and tape diagrams. Discussing the answers with the students and requiring them to justify their responses will help them draw connections among these representations and synthesize their understanding.

TEACHER MOVES

Engage the class in a discussion following the completion of the questions. Draw connections between the ways ratios are expressed and how they can be represented visually. Have students verbalize how they can use the ratio in one batch of cookies to determine the ratios for multiple batches of cookies to solidify understanding.

6.2.4 GUIDED INSTRUCTION: USING TAPE DIAGRAMS TO REPRESENT RATIOS

Component Overview: *Students use tape diagrams to represent ratios. Guide students through this component. Engage the class in a discussion to solidify the connection between the tape diagrams and ratios.*

DIFFERENTIATE INSTRUCTION

Consider using a kinesthetic representation of the number line on a clothesline. Students can place and adjust numbers on folded paper/cardstock on the clothesline.

ENRICHMENT

Students may struggle to understand how the same tape diagram can be used to represent a ratio comparing different numbers. Ask the following questions to clarify misconceptions:

- If the ratio of cats to dogs stays 4:5, how can you use the tape diagram to show this?
- How many cats and dogs were at the shelter in the first example?
- How can you determine this? How can you visually represent the new total of animals on the tape diagram? Consider how you can get from 9 to 27.

TEACHER MOVES

Students may struggle to conceptually understand and articulate why the ratios are equivalent, even if they are able to represent them correctly on the tape diagram.

Utilize inquiry ("How do you know?") for questions 2a-2e and talk moves ("Restate in your words what your classmate said.") to formalize their conceptual understanding before they independently practice in 6.2.5 Your Turn!.

6.2.4 Guided Instruction: Using Tape Diagrams to Represent Ratios

A tape diagram is another way to represent ratios. All the parts of the diagram that are the same size have the same value, similar to increments on a number line.

The tape diagram shown represents the ratio of cats to dogs at a local animal shelter.

number of cats					
number of dogs					

1. **Complete the statements using the tape diagram.**
   a. **The ratio of dogs to cats is** $\frac{5}{4}$.
   b. **For every** __4__ **cats at the shelter, there are** __5__ **dogs.**

2. **If there are 27 cats and dogs at the animal shelter in all, and the ratio of cats to dogs is 4:5, the same tape diagram can be used to represent the relationship.**

number of cats	3	3	3	3	
number of dogs	3	3	3	3	3

   a. **What is the value of each small rectangle in the diagram given this new information?** 3
   b. **Label each small rectangle with this value.** *Shown on diagram.*
   c. **How many dogs are at the animal shelter?** 15
   d. **How many cats are at the animal shelter?** 12
   e. **Complete the statement.** **The ratio of cats to dogs at the shelter, 4:5, is equivalent to the ratio** 12 **to** 15**.**

6.2.5 Your Turn!

1. Mr. Floyd is planning a field trip for his class to the Cade Museum for Creativity and Invention in Gainesville, Florida. To meet the museum's recommended ratio of chaperones to students, Mr. Floyd needs to have 1 adult chaperone for every 9 students. The tape diagram represents the ratio of adults to students needed.

number of adults	8								
number of students	8	8	8	8	8	8	8	8	8

   a. What is the ratio of students to chaperones required by the museum?

   Any form of ratio representation is accepted:
   9 to 1, 9:1 or $\frac{9}{1}$

   There are 72 students who will attend the trip from Mr. Floyd's classes. Use the tape diagram to determine how many chaperones will be needed.

   b. Label each small rectangle with the number of people it represents.
   Shown on diagram.

   c. Complete the statement.

   For all 72 students to attend, 8 chaperones are needed.

6.2.5 YOUR TURN!

Component Overview: *Students work independently to represent and interpret ratios using double number lines and tape diagrams. Students then connect equivalent ratios to multiplication and division equations more formally, creating equations to find equivalent ratios.*

ENRICHMENT

Students may struggle to use the tape diagram to represent the new ratio. Encourage students to refer back to the previous component and consider how they represented equivalent ratios in the previous examples.

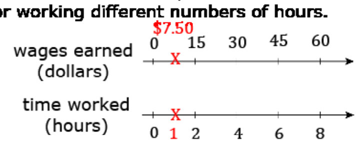
- If the ratio of students to teachers is 9:1, how many teachers will be needed for 72 students?
- How can you determine what this equivalent ratio will be?
- How did you determine the equivalent ratio in the previous example?

6.2.5 YOUR TURN! (CONTINUED)

TEACHER MOVES

Encourage students to think through how they can determine each amount and how they can show their work.

2. A cashier worked an 8-hour day and earned $60.00. The double number line represents the amount she earned for working different numbers of hours.



a. Determine the amount of money she earned in 1 hour. Show your work and label it on the double number line.

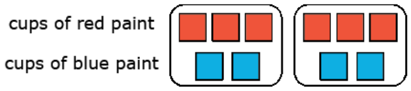
***Shown on diagram.* 60/8 = $7.50**

b. How much money does the cashier earn if she works 3 hours?

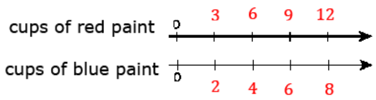
$7.50 x 3 = $22.50

6.2.6 Wrap-Up: Ticket Out the Door

1. The representation shows the amount of red and blue paint that makes 2 batches of a particular shade of purple paint called lilac.



   a. On the double number line, label the tick marks to represent amounts of red and blue paint used to make batches of this shade of purple paint.



   b. How many batches are made with 12 cups of red paint?
   4 batches

   c. How much red paint should be mixed with 6 cups of blue paint to make lilac paint?
   9 cups of red paint

   d. Complete the statement.

   To make this shade of lilac paint, mix 2 cups of blue paint for every 3 cups of red.

6.2.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Students use a tape diagram to create a double number line and interpret a ratio.*

DIFFERENTIATE INSTRUCTION

Students may struggle using a double number line but correctly apply ratios and identify solutions. There is not one “correct” way to represent ratios using various models as described in the benchmark.

For students who correctly answer questions 1b-1d but struggle with question 1a, consider offering them the opportunity to represent their solution using a model of their choice.